

Md. Tanvir Ahmed Tonmoy

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linkedin

Research Focus

VLSI Design, Compute-in-Memory, Memory-Based AI Accelerators, Emerging Memory Devices, Neuromorphic Computing.

Education

BRAC University, Dhaka, Bangladesh

(Expected September 2026)

Bachelor of Science in Electrical and Electronic Engineering

Specialization: Electronics

CGPA: 3.68/4.00

Publications

- H. H. Jui, **M. T. A. Tonmoy**, M. R. Kabbo, A. A. Mahi, N. M. Himel, S. Sultana, T. Titirsha, and T. Rahman, "Variation-Aware SRAM Compute-in-Memory Boolean Architecture for Binary Neural Networks," accepted in *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2026, Kolkata, India.
- F. Uddin, N. Shahrair, M. M. Rahman, **M. T. A. Tonmoy**, S. Akter, M. Jobayer, M. K. Rahman, and M. M. H. Shawon, "Soft Robotic Bellow-Controlled Arm Using Multi-DOF Control System," accepted in *5th Interdisciplinary Conference on Electrics and Computer (INTCEC) 2025*, Chicago, USA.
- M. M. Chowdhury, M. M. Rahman, M. Mahmud, K. A. Rahim, **M. T. A. Tonmoy**, A. Nawrin, C. D. Joy, R. S. I. Antara, and A. H. Kafi, "BRACU Diganta: An Affordable CanSat Learning Kit for Students in Developing Countries," accepted in *International Astronautical Congress (IAC/IAF)*, 2025, Sydney, Australia.

On-Going Work

- **SRAM-CIM MAC Accelerator.** (Current)
Developing a scalable SRAM-CIM MAC architecture based on compute-cell arrays and peripheral accumulation.
- **Verilog-A-Based RRAM Device and Neuromorphic Modeling.** (Current)
Exploring RRAM compact modeling, resistance-switching behavior, and conductance-state representation for future memory-centric and neuromorphic hardware primitives.

Research Experience

- **Compute-in-Memory.** (2026–Present)
Variation-Aware SRAM-CIM Boolean Architecture. Designed and simulated a 45 nm 6T SRAM-based compute-in-memory Boolean architecture in Cadence Virtuoso. Implemented Boolean operations including AND, OR, NAND, NOR, XOR, and XNOR, and performed Monte Carlo delay analysis to evaluate timing-induced logic error probability. Developed a hardware-aware BNN evaluation framework in Python/PyTorch to connect circuit-level timing variability with inference accuracy.
SRAM-CIM MAC Accelerator. Extending the Boolean CIM structure toward a scalable MAC accelerator by forming compute-cell arrays, integrating adder/accumulator structures, and evaluating delay, energy, and reliability trade-offs.
- **RRAM Device Modeling and Neuromorphic Hardware Exploration.** (2026–Present)
Verilog-A Based Resistive Memory Modeling. Currently studying resistive random-access memory (RRAM) device behavior and developing familiarity with Verilog-A based compact modeling for resistance switching, HRS/LRS states, and basic SET/RESET operation. Exploring how conductance-state behavior in RRAM devices can support future memory-centric and neuromorphic hardware primitives.
- **Multi DOF Soft Robotics Rehabilitation Device.** (2024–2025)
Pneumatic Bellows-Based Multi-DOF Rehabilitation Arm. Worked on a soft robotic rehabilitation system using industrial bellows as pneumatic actuators for multi-DOF motion. Integrated sensing and control concepts including pressure sensing, IMU feedback, and trajectory tracking for rehabilitation-oriented assistive robotics.
- **FYDP/Capstone Project.** (2026–Present)
Autonomous Agricultural Rover for Soil Sampling and Zone-Wise Fertilization. Developed a precision agriculture rover concept involving grid-based soil sampling, NPK analysis, zone-wise fertilization, and simulation-based evaluation in Webots/Gazebo. Worked on system design, sensor integration, motion planning, and performance evaluation.

Course and Other Projects

- **45 nm CMOS Phase-Locked Loop Design in Cadence.** (2026)
Designed and simulated a CMOS PLL using Cadence Virtuoso and Maestro, including PFD, charge pump, loop filter, VCO, and divide-by-48 frequency divider. Performed DC and transient analysis and evaluated locked-state behavior, current consumption, and frequency response.
- **BRACU Diganta CanSat and Model Satellite Systems.** (2024-Present)
Worked on onboard electronics and system development for BRACU Diganta, selected among the global finalists of CanSat 2025 in Virginia, USA. Designed and integrated sensor interfaces, telemetry electronics, beacon logic, and payload electronics. Also leading the electronics subsystem to Teknofest 2026 model satellite competition including science payload electronics, flight controller prototyping, communication subsystem planning, and system integration.
- **IoT-Based Air Quality Monitoring System.** (2026)
Developed an IoT-based environmental monitoring system for measuring air-quality related parameters using embedded sensors and microcontroller-based data acquisition. Worked on sensor interfacing, wireless data transmission, and real-time monitoring concepts for smart environmental sensing applications.
- **CNN-Based Floating Waste Classification on Water Surfaces.** (2026)
Implemented a CNN-based visual recognition pipeline for classifying floating waste materials on water surfaces into six categories: algae, chemical, mixed, natural, plastic, and polythene pollution. Worked on dataset preprocessing, image augmentation, CNN training, validation analysis, and qualitative testing for autonomous water-cleaning system applications.

Awards and Achievements

- **Merit Based Scholarship** – Awarded a 25% tuition scholarship for academic excellence for four semesters and 10% for three semesters.
- **Dean's List Award** – Recognized on the Dean's List for outstanding academic performance for six semesters.
- **CanSat Competition 2025 Global Finalist** – American Astronautical Society, Virginia, USA.
- **Industrial Attachment, Neural Semiconductor Ltd.** – Gained exposure to semiconductor design workflows including digital design, physical design, analog design, RF design, and fabrication-related processes.
- **Former Undergraduate Teaching Assistant** – Department of Electrical and Electronic Engineering, BRAC University.
- **International Conference Publications** – ISVLSI 2026, INTCEC 2025, and IAC/IAF 2025.

Leadership and Community Involvement

- **Electronics Subsystem Lead, BRACU Diganta CanSat Team** – Led onboard electronics design, sensor integration, power system planning, and telemetry-related development.
- **Research and Technical Contributor** – Contributed to interdisciplinary projects in VLSI, soft robotics, space systems, embedded systems, and precision agriculture.

References

- **Twisha Titirsha, PhD**
Assistant Professor, Department of Electrical and Electronic Engineering.
BSRM School Of Engineering, BRAC University.
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